Coordinate-free Mathematics

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The first section develops algebraic constructions independent of a basis; the second develops geometric constructions independent of coordinates; the third develops analytic constructions from norms, operators, and positivity.

1 Algebra

Linear algebra studies linear maps between vector spaces. A **vector space** V over a field k is a set equipped with vector addition and scalar multiplication operations satisfying straightforward compatibility conditions. A **linear map** $f : V \to U$ between k-vector spaces is a map of underlying sets such that $f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$ for all $\alpha, \beta \in k$ and $v, w \in V$. Examples include differential operators, group representations, coefficient matrices of linear systems of equations, and local approximations of globally nonlinear functions [1, 2].

The **free k-vector space on a set** X, denoted kX, is the set of finitely supported set-theoretical functions $X \to k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W, denoted V/W, is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is the vector space $k(V \times W)/N$ where N is the subspace generated by the obvious linearity relations on V and W.

An **isomorphism** of vector spaces $V \cong W$ consists of linear maps $f : V \to W$ and $g : W \to V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$.

Exercise 1 (Universal constructions)

Prove the following:

- $k(X \sqcup Y) \cong kX \oplus kY$
- $k(X \times Y) \cong kX \otimes kY$
- *The first isomorphism theorem for vector spaces (and second and third if you want).*

- *Find universal properties determining up to canonical isomorphism the free k-vector space, direct sum, quotient, tensor product, and dual space (defined below), respectively.*

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k. The evaluation map $\mathrm{ev}_V : V^* \otimes V \to k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\mathrm{coev}_V : k \to V \otimes V^*$ satisfying the snake equations $(1_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes 1_V) = 1_V$ and $(\mathrm{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \mathrm{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \mathrm{tr}(1_V)$, where $\mathrm{tr}(f) : \mathrm{End}(V) \to k$ is given by $f \mapsto (\mathrm{ev}_V \circ s_{V,V^*} \circ (f \otimes 1_{V^*}) \circ \mathrm{coev}_V)(1)$ where $s_{X,Y} : X \otimes Y \to Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$ [7].

Exercise 2 (Trace and dimension identities)
Prove the following for finite-dimensional vector spaces V, W and linear maps $f, g : V \to V$.

- $\mathrm{tr}(fg) = \mathrm{tr}(gf)$
- $\mathrm{tr}(f \oplus g) = \mathrm{tr}(f) + \mathrm{tr}(g)$
- $\mathrm{tr}(f \otimes g) = \mathrm{tr}(f)\,\mathrm{tr}(g)$
- $\dim(V) \in \mathbf{N}$
- $V \cong W$ *iff* $\dim(V) = \dim(W)$

Define the tensor algebra and **exterior algebra of** V by [4]

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree-k component of ΛV.

Exercise 3 (Top exterior power) *Show that the highest graded component of ΛV is one-dimensional.*

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots f(v_{\dim(V)})$.

For an n-dimensional real vector space V, its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive.

Exercise 4 (Determinant identities and volume)
Prove the following:

- $\det(f \oplus g) = \det(f)\det(g)$
- *If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0, 1]$.*

- *Cramer's rule*

For $f \in \mathrm{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ.

Exercise 5 (Characteristic polynomial and Jordan form)

Let $f \in \mathrm{End}(V)$ with V finite dimensional. Suppose its characteristic polynomial splits over k. Prove:

- *Every eigenvalue of f is a root of its characteristic polynomial.*
- *The geometric multiplicity of each eigenvalue is at most equal to its **algebraic multiplicity** (i.e. multiplicity as a root of the characteristic polynomial)*
- $\mathrm{tr}(f) = \sum_i \lambda_i$ *and* $\det(f) = \prod_i \lambda_i$*, where each eigenvalue λ_i is repeated according to its algebraic multiplicity*
- *The operator f satisfies its own characteristic polynomial (replacing λ with f)*
- *The existence of the Jordan normal form for a linear operator on a finite-dimensional complex vector space*

A **Hermitian inner product** on a complex vector space V is a positive-definite sesquilinear form $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable and linear in the second. It induces the norm $\|v\| = \sqrt{\langle v, v \rangle}$. For a linear map $A : V \to W$ between finite-dimensional Hermitian spaces, an **adjoint** is a map $A^* : W \to V$ satisfying

$$\langle Av, w \rangle = \langle v, A^* w \rangle.$$

An endomorphism A is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle v, Av \rangle \geq 0$ for every v, and **unitary** if $A^*A = AA^* = 1$. Equip $\mathrm{End}(V)$ with the induced operator norm.

Exercise 6 (Finite-dimensional spectral theorem)

Prove that every linear map between finite-dimensional Hermitian spaces has a unique adjoint and establish the identities $(AB)^* = B^*A^*$ and $\|A^*A\| = \|A\|^2$. Prove that $A \in \mathrm{End}(V)$ is positive if and only if $A = B^*B$ for some B, and prove that every normal A has an orthogonal spectral decomposition

$$A = \sum_\lambda \lambda p_\lambda, \qquad p_\lambda p_\mu = \delta_{\lambda\mu} p_\lambda, \qquad \sum_\lambda p_\lambda = 1.$$

Apply the theorem to T^*T to prove that every map $T : H \to K$ between finite-dimensional Hermitian spaces has a singular-value decomposition

$$T = U \Sigma V^*.$$

Prove that the diagonal entries of Σ are the square roots of the nonzero eigenvalues of T^*T, counted with multiplicity.

1.1 Group representations

An action of a group G on a set X assigns to each $g \in G$ a bijection $x \mapsto gx$ such that $1x = x$ and $(gh)x = g(hx)$. A **representation** of G on a vector space V is a homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

A subspace $W \subseteq V$ is **G-invariant** if $\rho(g)W \subseteq W$ for every $g \in G$. Restriction gives the **subrepresentation** W, and $\rho(g)(v + W) = \rho(g)v + W$ gives the **quotient representation** V/W. A nonzero representation is **irreducible** if its only invariant subspaces are 0 and V.

For representations V and W, the direct sum, tensor product, and dual representations are defined by

$$\begin{aligned} g(v, w) &= (gv, gw), \\ g(v \otimes w) &= gv \otimes gw, \\ (g\phi)(v) &= \phi(g^{-1}v), \qquad \phi \in V^*. \end{aligned}$$

An **intertwiner** $T : V \to W$ is a linear map satisfying $T\rho_V(g) = \rho_W(g)T$ for every g. Write $\mathrm{Hom}_G(V, W)$ for the space of intertwiners and

$$V^G = \{v \in V : \rho(g)v = v \text{ for every } g \in G\}$$

for the fixed-point space [5].

Exercise 7 (Invariant averaging)

Let G be finite and let V be a complex representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

Prove that P_G is the projection of V onto V^G. Prove also that averaging any Hermitian inner product over G produces a G-invariant Hermitian inner product, so every finite-dimensional complex representation of G is equivalent to a unitary representation.

1.2 Maschke's theorem and Schur's lemma

Exercise 8 (Maschke's theorem)

Let $W \subseteq V$ be G-invariant and let $P : V \to W$ be any linear projection. Define

$$\widetilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}.$$

Prove that $\widetilde{P}$ is G-equivariant and has image W. Deduce that $\ker \widetilde{P}$ is a G-invariant complement of W. If G is finite and the characteristic of k does not divide $|G|$, prove that every finite-dimensional k-representation of G is a direct sum of irreducible representations.

Exercise 9 (Schur's lemma)

Let V and W be irreducible finite-dimensional complex representations. Prove that every nonzero intertwiner $T : V \to W$ is an isomorphism and that

$$\mathrm{End}_G(V) = \mathbb{C}1_V.$$

Let $\mathbb{C}[G]$ act on V by the linear extension of ρ. Deduce that every element of the center $Z(\mathbb{C}[G])$ acts by a scalar on each irreducible representation.

1.3 Characters and orthogonality

The **character** of a finite-dimensional representation V is the function

$$\chi_V(g) = \mathrm{tr}(\rho_V(g)).$$

For class functions $\chi, \psi : G \to \mathbb{C}$, define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 10 (Great Orthogonality Theorem)

Prove that characters are class functions and that, for unitary representations,

$$\chi_{V \oplus W} = \chi_V + \chi_W, \qquad \chi_{V \otimes W} = \chi_V \chi_W, \qquad \chi_{V^*}(g) = \overline{\chi_V(g)}.$$

Let Γ^α and Γ^β be irreducible unitary representations of a finite group G, of dimensions d_α and d_β. In orthonormal bases, prove that

$$\sum_{g \in G} \Gamma^\alpha(g)_{ij} \overline{\Gamma^\beta(g)_{k\ell}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{j\ell}.$$

Deduce that their characters satisfy

$$\langle \chi_\alpha, \chi_\beta \rangle_G = \delta_{\alpha\beta}.$$

If V is a finite-dimensional representation, prove that the multiplicity of V_α in V is

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G.$$

Conclude that the character determines a finite-dimensional complex representation up to isomorphism.

For an irreducible character χ_α, define the **symmetry projection**

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g).$$

Exercise 11 (Symmetry projections)

Prove that the operators P_α are pairwise orthogonal idempotents and that $P_\alpha V$ is the α-isotypic component of V. Interpret P_α as constructing symmetry-adapted linear combinations of atomic orbitals or displacement vectors.

1.4 Molecular symmetry

The **point group** of a molecule is the finite group $G \subseteq O(3)$ of spatial symmetries preserving its nuclear configuration, including atomic species. For a molecule with N atoms, G acts on the real displacement space

$$V_{\text{disp}} = \bigoplus_{a=1}^{N} \mathbb{R}^3_a \cong \mathbb{R}^{3N}$$

by permuting atoms and applying the spatial linear action to each displacement. The three-dimensional translational representation carries the polar-vector action R_g, while the rotational representation carries the axial-vector action $\det(R_g)R_g$; denote them by V_{trans} and V_{rot}. We complexify these real representations when applying character theory. For a nonlinear molecule, the vibrational representation is the class

$$[V_{\text{vib}}] = [V_{\text{disp}}] - [V_{\text{trans}}] - [V_{\text{rot}}]$$

in the representation ring [6].

Exercise 12 (Character of molecular displacements)
For $g \in G$, prove that

$$\chi_{\text{disp}}(g) = \sum_{\substack{a \\ g(a)=a}} \operatorname{tr}(R_g),$$

where $R_g : \mathbb{R}^3 \to \mathbb{R}^3$ is the spatial action and the sum is over atoms fixed by g. Prove that $\dim V_{\text{vib}} = 3N - 6$ for a nonlinear molecule, and use character orthogonality to express the multiplicity of every irreducible representation in V_{vib}.

For water, take the z-axis along the C_2 axis and the molecule in the yz-plane. Its point group is

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Exercise 13 (Vibrations of H_2O)
Use the group law and character orthogonality to derive the character table

	E	C_2	σ_{xz}	σ_{yz}	*basis functions*
A_1	1	1	1	1	z
A_2	1	1	-1	-1	R_z
B_1	1	-1	1	-1	$x,\ R_y$
B_2	1	-1	-1	1	$y,\ R_x$.

Compute

$$\chi_{\text{disp}} = (9, -1, 1, 3), \qquad \chi_{\text{trans}} = (3, -1, 1, 1), \qquad \chi_{\text{rot}} = (3, -1, -1, -1).$$

Use character orthogonality to prove that

$$V_{\mathrm{vib}} \cong 2A_1 \oplus B_2.$$

Identify the two A_1 modes as the symmetric stretch and bend, and the B_2 mode as the asymmetric stretch. Construct symmetry-adapted displacement vectors using the projections P_{A_1} and P_{B_2}.

1.5 Spectroscopic selection rules

Let V_i and V_f be the symmetry types of initial and final states. The electric-dipole operator transforms as the vector representation $V_\mu = V_{\mathrm{trans}}$. A transition amplitude

$$\langle \psi_f, \mu \psi_i \rangle$$

is the matrix coefficient used below.

Exercise 14 (Electric-dipole selection rule)
Use the Great Orthogonality Theorem to prove that the transition amplitude vanishes unless

$$1 \subseteq V_f^* \otimes V_\mu \otimes V_i.$$

For irreducible V_f, prove that this is equivalent to

$$V_f \subseteq V_\mu \otimes V_i.$$

The vibrational ground state of water has type A_1. Since

$$V_\mu \cong A_1 \oplus B_1 \oplus B_2,$$

deduce that both A_1 modes and the B_2 mode are infrared active. Thus molecular symmetry alone predicts that all three fundamental vibrations of water are allowed in infrared spectroscopy.

Exercise 15 (Optional Raman selection rule)
Let the polarizability tensor carry $\mathrm{Sym}^2(V_\mu)$. Prove that a vibrational mode is Raman active precisely when its irreducible representation occurs in this symmetric square. Prove that

$$\mathrm{Sym}^2(V_\mu) \cong 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$$

for C_{2v}. Deduce that all three fundamental vibrational modes of water are Raman active.

Finite-group representations turn molecular symmetry into experimentally testable selection rules. The monoidal categories below retain irreducible decomposition and tensor product as fusion rules, and add associators and braiding.

1.6 Monoidal and braided categories

A $\mathbb{C}$-linear category is a category whose morphism sets

$$\mathrm{Hom}_{\mathcal{C}}(x, y)$$

are complex vector spaces and whose composition is bilinear.

A $\mathbb{C}$-linear monoidal category is a $\mathbb{C}$-linear category equipped with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1, up to coherent natural isomorphism. Direct sums and duals generalize the corresponding vector-space constructions.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\mathrm{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities [35].

The definitions of duality, trace, and dimension above extend verbatim to $\mathcal{C}$. Write $d_x = \mathrm{tr}(1_x)$. Assume that the categorical dimensions of nonzero objects are positive real numbers.

Exercise 16 (Categorical trace and dimension)
Repeat the first three parts of Exercise 2 in $\mathcal{C}$.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \to x'$ and $g : y \to y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$c_{x,y\otimes z} = (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z),$$
$$c_{x\otimes y,z} = (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}).$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\mathrm{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$. Assume first that every simple object is invertible and that their isomorphism classes form the cyclic group

$$\{1, a, a^2, \ldots, a^{m-1}\}, \qquad a^r \otimes a^s \cong a^{r+s \pmod m}.$$

Write $U(1) = \{z \in \mathbb{C} : |z| = 1\}$ and $M_{r,s} = M_{a^r,a^s} \in U(1)$.

Exercise 17 (Abelian braiding phase)
Use the hexagon identities to prove that

$$M_{r+s,t} = M_{r,t}M_{s,t}, \qquad M_{r,s+t} = M_{r,s}M_{r,t}.$$

Deduce that there is an integer k modulo m such that

$$M_{r,s} = \exp\left(\frac{2\pi ikrs}{m}\right).$$

For $k = 1$, prove that the monodromy of a about N copies of a is multiplication by

$$\exp\left(\frac{2\pi iN}{m}\right).$$

In particular, for $m = 3$, increasing N by one changes the phase by $2\pi/3$.

The phase $2\pi/3$ is the value observed by interferometry in the $\nu = 1/3$ fractional quantum Hall state [37].

Now suppose that $\mathcal{C}$ has two simple objects $1, \tau$ satisfying the fusion rule

$$\tau \otimes \tau \cong 1 \oplus \tau.$$

Exercise 18 (Fibonacci categorical dimension)
Let $d_\tau = \mathrm{tr}(1_\tau)$. Prove that [35]

$$d_\tau^2 = 1 + d_\tau$$

and, using positivity, that

$$d_\tau = \frac{1+\sqrt{5}}{2}.$$

Let

$$V_n = \mathrm{Hom}_{\mathcal{C}}(1, \tau^{\otimes n}).$$

Show that the dimensions of these fusion spaces satisfy the Fibonacci recurrence and determine their asymptotic growth rate.

Let

$$V = \mathrm{Hom}_{\mathcal{C}}(\tau, \tau^{\otimes 3}).$$

Set $\varphi = (1+\sqrt{5})/2$. In the basis indexed by the intermediate labels $1, \tau$, define [36]

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \qquad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

Define $\sigma_1 = R$ and $\sigma_2 = F^{-1}RF$. Let

$$B_3 = \langle b_1, b_2 \mid b_1b_2b_1 = b_2b_1b_2 \rangle.$$

Exercise 19 (Non-Abelian braid representation)

Use the fusion rule to prove that V is two-dimensional with basis indexed by the intermediate objects $1, \tau$ in $(\tau \otimes \tau) \otimes \tau$. Working exactly, verify that F and R are unitary and prove that

$$\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2.$$

Prove also that $\sigma_1\sigma_2 \neq \sigma_2\sigma_1$. Deduce that $b_i \mapsto \sigma_i$ defines a non-Abelian representation

$$B_3 \longrightarrow U(V).$$

Such Fibonacci fusion and braiding data have been implemented on superconducting quantum processors [39].[1]

2 Geometry

2.1 Differential forms and cohomology

A **smooth manifold** is a topological space equipped with an atlas whose transition maps are smooth. A smooth map $f : M \to N$ has a tangent map $df : TM \to TN$. The **tangent bundle** TM is the union of the tangent spaces, the **cotangent bundle** is T^*M, and a **vector field** is a smooth section of TM [3]. An orientation of M is a smoothly varying choice of orientation of its tangent spaces, using the determinant lines defined in Algebra.

The exterior powers $\Lambda^k T_p^* M$ are those defined in the Algebra section. To define integration in a coordinate-free way, we will integrate differential forms over chains. The space of **differential n-forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^*M)$. On an oriented n-manifold, a **volume form** is a nowhere-vanishing element of $\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \to \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies [8]

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n-simplex** is a smooth map $\sigma : \Delta^n \to M$ where Δ^n is the standard n-simplex. The collection of singular n-simplices in M freely generates the abelian group $C_n(M)$ of **singular n-chains** in M. The **boundary map** $\partial : C_n(M) \to C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \to M$ by

$$\partial\sigma = \sum_{j=0}^{n} (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \to \Delta^n$ is the inclusion of the face opposite the j-th vertex of Δ^n.

[1] This is an engineered quantum simulation of Fibonacci topological order, not evidence that naturally occurring Fibonacci quasiparticles have been conclusively observed. A related trapped-ion realization of non-Abelian topological order is reported in [38].

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \to A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n. The n-th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n)/\operatorname{im}(d_{n+1}).$$

The prototypical example is singular homology which corresponds to the chain complex of singular chains with the boundary map defined above.

Cohomology and cochains are defined similarly, using superscripts instead of subscripts and boundary maps $d^n : A^n \to A^{n+1}$ increase degree instead of decrease it. Singular cohomology corresponds to the cochain complex given by dualizing the singular chain complex above.

The **de Rham cohomology** of M is

$$H^n_{\mathrm{dR}}(M) = \ker(d \colon \Omega^n(M) \to \Omega^{n+1}(M))/\operatorname{im}(d \colon \Omega^{n-1}(M) \to \Omega^n(M)).$$

For a smooth map $f : N \to M$, the **pullback** of $\alpha \in \Omega^p(M)$ is

$$(f^*\alpha)_x(v_1, \dots, v_p) = \alpha_{f(x)}(df_x v_1, \dots, df_x v_p).$$

Exercise 20 (Poincaré lemma and Stokes' theorem)

Prove that $(f \circ g)^* = g^* f^*$, that pullback preserves wedge products, and that $d(f^*\alpha) = f^*(d\alpha)$. Prove that $d^2 = 0$. Prove the Poincaré lemma: every closed form is locally exact in positive degree. Define the integral of an n-form on an n-chain and prove

$$\int_c d\omega = \int_{\partial c} \omega.$$

Exercise 21 (de Rham theorem)

Prove that integration over smooth singular cycles induces a natural isomorphism between de Rham cohomology and singular cohomology with real coefficients.

The Aharonov–Bohm effect gives a direct physical meaning to first de Rham cohomology. In the region accessible to a charged particle outside an impenetrable solenoid, the electromagnetic potential A may satisfy $dA = 0$ without being globally exact. Its periods then produce observable quantum phases even though the magnetic field vanishes along every particle trajectory [34, 26].

Exercise 22 (Aharonov–Bohm effect and de Rham cohomology)

Let M be a smooth manifold and let $A \in \Omega^1(M)$ satisfy $dA = 0$. Prove the following:

- *For every contractible open set $U \subseteq M$, there exists $\lambda \in C^\infty(U)$ such that $A|_U = d\lambda$.*
- *For every smooth closed curve γ, the period $\int_\gamma A$ depends only on the homology class of γ.*

- *There exists a global $\lambda \in C^\infty(M)$ with $A = d\lambda$ if and only if $[A] = 0 \in H^1_{\mathrm{dR}}(M)$. If $[A] \neq 0$, use the de Rham theorem to find a smooth closed curve with nonzero period.*

Now take $M = \mathbb{R}^3 \setminus \{(0,0,z) : z \in \mathbb{R}\}$, the space outside an idealized solenoid, and define

$$d\theta = \frac{-y\,dx + x\,dy}{x^2 + y^2}, \qquad A = \frac{\Phi}{2\pi} d\theta.$$

Prove that M deformation retracts onto S^1, that $d\theta$ is closed but not exact, and that its integral around a positively oriented loop linking the solenoid once is 2π. Deduce that $\int_\gamma A = \Phi$. For a particle of charge q, prove that

$$\exp\left(\frac{iq}{\hbar}\int_\gamma A\right)$$

is unchanged by $A \mapsto A + d\lambda$, and explain why the relative phase $q\Phi/\hbar$ shifts the interference pattern although $dA = 0$ on M.

Exercise 23 (Categorical trace on chain complexes)
Prove that bounded chain complexes of finite-dimensional vector spaces form a symmetric monoidal abelian category with duals, that the categorical trace of an endomorphism is its Lefschetz number, and that the categorical dimension of a chain complex is its Euler characteristic.

2.2 Hodge theory and electromagnetism

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n-manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. A **metric volume form** vol_g is a positive volume form satisfying

$$|\langle \mathrm{vol}_g, \mathrm{vol}_g \rangle_g| = 1.$$

The **Hodge star** is an operator $\star : \Omega^p(M) \to \Omega^{n-p}(M)$ satisfying

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \,\mathrm{vol}_g\,.$$

Exercise 24 (Metric volume and Hodge star)
Prove that an oriented pseudo-Riemannian manifold has a unique metric volume form and a unique Hodge-star operator. Determine $\star^2$ in terms of the degree and signature of the metric.

On an oriented Lorentzian four-manifold, the electromagnetic field strength is a two-form F and the electric four-current is a three-form J. Maxwell's equations are [26]

$$dF = 0, \qquad d \star F = J.$$

Exercise 25 (Maxwell's equations in spacetime)

Prove that Maxwell's equations imply $dJ = 0$. If C is an oriented compact four-dimensional chain, deduce that

$$\int_{\partial C} J = 0.$$

If the three-dimensional pieces of its boundary satisfy

$$\partial C = \Sigma_1 - \Sigma_0 + B,$$

prove that

$$\int_{\Sigma_1} J - \int_{\Sigma_0} J = -\int_B J.$$

Specialize to Minkowski spacetime $\mathbb{R}_t \times \mathbb{R}^3$ with metric $-dt^2 + dx^2 + dy^2 + dz^2$. Let $\star_3$ and vol_3 be the Euclidean Hodge star and volume form on space, and write

$$F = B - dt \wedge E, \qquad J = \rho\, \mathrm{vol}_3 - dt \wedge \star_3 j,$$

where E, j are spatial one-forms and B is a spatial two-form. Prove that Maxwell's equations are equivalent to

$$d_3 B = 0, \qquad \partial_t B + d_3 E = 0,$$

$$d_3 \star_3 E = \rho\, \mathrm{vol}_3, \qquad d_3 \star_3 B - \partial_t \star_3 E = \star_3 j.$$

Identify these with Gauss's laws, Faraday's law, and the Ampère–Maxwell law. Show that $dJ = 0$ becomes the continuity equation

$$\partial_t \rho\, \mathrm{vol}_3 + d_3 \star_3 j = 0.$$

Exercise 26 (Maxwell action and gauge invariance)

Assume that M has no boundary, or restrict to compactly supported variations. For $A \in \Omega^1(M)$, set $F = dA$ and define

$$S[A] = -\frac{1}{2} \int_M F \wedge \star F + \int_M A \wedge J.$$

For $\alpha \in \Omega^1(M)$, compute the first variation of $S[A + t\alpha]$ and prove that its Euler–Lagrange equation is $d \star F = J$. Explain why $dF = 0$ holds identically. Prove that the action is invariant under $A \mapsto A + d\lambda$ when $dJ = 0$, and show conversely how gauge invariance of the source term forces charge conservation.

2.3 Bundles, K-theory, and band topology

A **smooth vector bundle** $p : E \to M$ is a smooth family of vector spaces locally isomorphic to a product $U \times V \to U$. For a smooth map $f : N \to M$, its **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \longrightarrow N.$$

For a compact Hausdorff space X, let $\mathrm{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X, with the usual direct sum, tensor product, dual, and pullback constructions [9]. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E \oplus F] = [E] + [F].$$

Define multiplication by $[E][F] = [E \otimes F]$. For a continuous map $f : X \to Y$ between compact Hausdorff spaces, define pullback by

$$f^* : K^0(Y) \longrightarrow K^0(X), \qquad [E] \longmapsto [f^*E].$$

Exercise 27 (Grothendieck group of vector bundles)

Prove that $(f \circ g)^*E \cong g^*f^*E$ and that pullback preserves direct sums, tensor products, and duals. Let X and Y be compact Hausdorff spaces and let $f : X \to Y$ be continuous. Prove that direct sum and tensor product induce a commutative ring structure on $K^0(X)$ and that f^* is a ring homomorphism. Prove that

$$K^0(\mathrm{pt}) \cong \mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0 \in X$, regarded as a map $x_0 : \mathrm{pt} \to X$. The **reduced K-theory** of the pointed space (X, x_0) is

$$\widetilde{K}^0(X) = \ker\left(K^0(X) \xrightarrow{x_0^*} K^0(\mathrm{pt})\right).$$

For a bundle E of rank r, its **reduced class** is

$$[E] - [\underline{\mathbb{C}}^r],$$

where $\underline{\mathbb{C}}^r = X \times \mathbb{C}^r$ is the trivial rank-r bundle. Write $K^n(X)$ for the generalized cohomology groups extending $K^0(X)$.

Exercise 28 (Stable equivalence and Bott periodicity)

Let E and F be complex vector bundles over a compact Hausdorff space X. Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \underline{\mathbb{C}}^n$. Prove Bott periodicity

$$K^{n+2}(X) \cong K^n(X)$$

and deduce that complex K-theory is naturally $\mathbb{Z}/2$-graded.

For the gapped families below, assume that X is connected, so vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space as in Algebra. Exercise 6 supplies the spectral decomposition used below. A continuous family is a map

$$H : X \longrightarrow \mathrm{End}(V)$$

and write $H_x = H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Given a positively oriented contour Γ enclosing the negative spectra and excluding the positive spectra, define

$$P_x^- = \frac{1}{2\pi i}\int_\Gamma (z - H_x)^{-1}\, dz, \qquad E_x^- = \mathrm{im}(P_x^-).$$

Exercise 29 (Negative spectral bundle)
Let H be a gapped family over compact X. Prove that there are $\delta, M > 0$ such that every eigenvalue of every H_x lies in $[-M, -\delta] \cup [\delta, M]$, and hence that a contour Γ as above exists. Prove that

$$(P_x^-)^2 = P_x^-, \qquad (P_x^-)^* = P_x^-,$$

that $x \mapsto P_x^-$ is continuous, and that

$$E_H^- = \coprod_{x\in X} E_x^-$$

is a complex vector bundle over X. Prove that its rank is locally constant.

For a gapped family $H : X\times[0,1] \to \mathrm{End}(V)$, let E_j^- be the negative spectral bundle of $x \mapsto H(x, j)$, and let $i_j : X \to X \times [0, 1]$ be given by $i_j(x) = (x, j)$.

Exercise 30 (Gap-preserving homotopy)
Prove that the negative spectral subspaces of H form a vector bundle $E^- \to X \times [0,1]$. Prove that the endpoint pullbacks of any vector bundle over $X \times [0, 1]$ are isomorphic, and apply this to $i_0^*E^-$ and $i_1^*E^-$. Deduce that

$$[E_0^-] = [E_1^-] \in K^0(X).$$

Let $Q \in \mathrm{End}(W)$ be a fixed invertible self-adjoint endomorphism of a finite-dimensional Hermitian space W, let W^- be its negative eigenspace, and write $\underline{W^-} = X \times W^-$.

Exercise 31 (Stabilization)
For a gapped family H, prove that

$$E_{H\oplus Q}^- \cong E_H^- \oplus \underline{W^-}.$$

Deduce that

$$[E_H^-] - [\underline{\mathbb{C}}^{\,\mathrm{rank}\, E_H^-}] \in \widetilde{K}^0(X)$$

is unchanged by adjoining a constant gapped summand.

Let $E \subseteq X \times V$ be a subbundle and let P_x be the orthogonal projection onto E_x.

Exercise 32 (Flattened family)
Define $H_x = 1_V - 2P_x$. Prove that $x \mapsto H_x$ is a gapped family, that

$$H_x^2 = 1_V,$$

and that its negative spectral bundle is E. Deduce that every subbundle of a trivial bundle arises as the negative spectral bundle of a gapped family. Combine the preceding exercises to prove that a finite-dimensional gapped family determines a class in $\widetilde{K}^0(X)$ invariant under gap-preserving homotopy and stabilization.

Consider a translation-invariant noninteracting band insulator in symmetry class A: there is no time-reversal, particle-hole, or chiral symmetry, and its Bloch Hamiltonian is a smooth gapped family

$$H : \mathbb{T}^d \longrightarrow \mathrm{End}(V)$$

over the Brillouin torus. The negative-energy bundle E_H^- is the bundle of occupied states. Declare two such Hamiltonians **stably equivalent** when they become connected by a smooth gap-preserving homotopy after adjoining momentum-independent filled or empty bands [10].

Exercise 33 (Class A topological band insulators)
Use the preceding exercises and the stable classification of vector bundles by homotopy classes of projections to prove that

$$H \longmapsto [E_H^-] - [\underline{\mathbb{C}}^{\operatorname{rank} E_H^-}]$$

induces a bijection from stable equivalence classes of class A Hamiltonians over a compact connected space X to $\widetilde{K}^0(X)$. In particular, show that every reduced K-theory class is realized by a flattened Hamiltonian $H = 1 - 2P$.

Using the suspension isomorphism and Bott periodicity, prove that the strong class A phases of continuum models whose momentum space compactifies to S^d form

$$\widetilde{K}^0(S^d) \cong \begin{cases} \mathbb{Z}, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Using the Künneth theorem and the graded ring $K^*(S^1)$, prove that

$$K^0(\mathbb{T}^d) \cong \bigoplus_{\substack{0 \le j \le d \\ j \text{ even}}} \mathbb{Z}^{\binom{d}{j}}, \qquad \widetilde{K}^0(\mathbb{T}^d) \cong \bigoplus_{\substack{2 \le j \le d \\ j \text{ even}}} \mathbb{Z}^{\binom{d}{j}}.$$

Deduce that the reduced classifications over $\mathbb{T}^d$ for $d = 1, 2, 3$ are respectively 0, $\mathbb{Z}$, and $\mathbb{Z}^3$. Distinguish the strong invariant from the weak invariants inherited from lower-dimensional subtori.

This exercise does not give the full tenfold-way classification. Antiunitary symmetries impose additional Real or Quaternionic structure on the occupied states and require corresponding variants of K-theory [10].

2.4 Symplectic geometry

For a vector field X, contraction is the map $\iota_X : \Omega^p(M) \to \Omega^{p-1}(M)$ defined by

$$(\iota_X \alpha)(X_2, \ldots, X_p) = \alpha(X, X_2, \ldots, X_p).$$

The Lie derivative of forms is determined by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X \omega$ is a bundle isomorphism $TM \to T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M. For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H} \omega = dH.$$

Exercise 34 (Liouville's theorem)

Let (M, ω) be $2n$-dimensional with ω symplectic. Prove that $\omega^n/n!$ is a volume form. For $H \in C^\infty(M)$, prove using Cartan's formula that [24]

$$\mathcal{L}_{X_H} \omega = 0.$$

Deduce that $\mathcal{L}_{X_H} \omega^n = 0$ and hence that the local flow of X_H preserves the volume form

$$\frac{\omega^n}{n!}.$$

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

The following is a Hamiltonian symmetry–conservation statement, not the full variational theorem of [25].

Exercise 35 (Hamiltonian Noether theorem)

Let (M, ω) be symplectic. Prove that [24]

$$X_g(f) = \{f, g\}.$$

Let Φ_t be the local flow of X_f. Prove that the following are equivalent:

- *$H \circ \Phi_t = H$ whenever both sides are defined.*
- *$X_f(H) = 0$.*
- *$\{H, f\} = 0$.*
- *f is constant along the integral curves of X_H.*

2.5 Connections and curvature

An **affine connection** is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$ such that

$$\nabla_{fX} Y = f \nabla_X Y, \qquad \nabla_X (fY) = X(f)Y + f \nabla_X Y.$$

For a covector field α, set

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

Define the covariant derivative of general tensor fields by the product rule and by requiring it to commute with contractions.

A connection is **torsion-free and metric-compatible** when

$$\nabla_X Y - \nabla_Y X = [X, Y], \qquad \nabla g = 0.$$

Such a connection is called a **Levi-Civita connection**.

Exercise 36 (Levi-Civita connection)

Prove that every pseudo-Riemannian manifold admits a unique Levi-Civita connection. Derive the coordinate-free Koszul formula.

An affinely parametrized curve $\gamma : I \to M$ is a **geodesic** if $\nabla_{\dot\gamma} \dot\gamma = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_X g)(Y, Z) = X(g(Y, Z)) - g([X, Y], Z) - g(Y, [X, Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_X g = 0$.

Let (M, g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇ [27, 28].

Exercise 37 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt} g(\dot\gamma, \dot\gamma) = 0, \qquad \frac{d}{dt} g(X, \dot\gamma) = 0.$$

Deduce that $g(X, \dot\gamma)$ is constant along γ.

The **curvature** of ∇ is the map

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Exercise 38 (Curvature as a tensor)

For $f \in C^\infty(M)$ and vector fields X, Y, Z, prove that

$$R(fX, Y)Z = fR(X, Y)Z, \qquad R(X, fY)Z = fR(X, Y)Z,$$

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

Prove that $R(X, Y) = -R(Y, X)$ and deduce that

$$R \in \Gamma\left(\Lambda^2 T^* M \otimes T^* M \otimes TM\right).$$

Let $\Gamma : (-\varepsilon, \varepsilon) \times I \to M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by

$$T = \frac{\partial \Gamma}{\partial t}, \qquad J = \frac{\partial \Gamma}{\partial s}.$$

Exercise 39 (Geodesic deviation) *For a variation Γ as above, prove that $[T, J] = 0$ and that*

$$\nabla_T \nabla_T J = R(T, J)T.$$

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\operatorname{Ric}(X, Y) = \operatorname{tr}\big(Z \mapsto R(Z, X)Y\big), \qquad S = \operatorname{tr}_g(\operatorname{Ric}), \qquad G = \operatorname{Ric} - \frac{1}{2} S g.$$

For a symmetric covariant two-tensor A, its **divergence** is the covector field

$$(\operatorname{div} A)(Y) = \operatorname{tr}_g\big((X, Z) \mapsto (\nabla_X A)(Z, Y)\big).$$

Exercise 40 (Contracted Bianchi identity and conservation)
Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\operatorname{div} G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\operatorname{div} T = 0$*.*

Connections and curvature also determine characteristic classes. In odd dimensions, their transgression forms define Chern–Simons functionals.

Let $E \to M$ be a complex vector bundle and write $\Omega^0(M, E) = \Gamma(E)$. A **connection** on E is a $\mathbb{C}$-linear map

$$\nabla : \Omega^0(M, E) \longrightarrow \Omega^1(M, E)$$

satisfying $\nabla(fs) = df \otimes s + f \nabla s$. For $\alpha \in \Omega^p(M)$ and an E-valued form η, extend it by

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^p \alpha \wedge \nabla \eta.$$

Its **curvature** is

$$F_\nabla = \nabla^2 \in \Omega^2(M, \operatorname{End} E).$$

The connection induces one on $\operatorname{End} E$-valued forms.

Exercise 41 (Bianchi identity) *Let ∇ be a connection on $E \to M$. Prove that*
$$\nabla F_\nabla = 0.$$

Let $H : X \to \mathrm{End}(V)$ be a smooth family of self-adjoint operators with a simple isolated eigenvalue λ_x, and set

$$L = \{(x, v) \in X \times V : H_x v = \lambda_x v\}.$$

For a local unit eigenvector ψ, define the **Berry connection** and **Berry curvature** by
$$\mathcal{A} = i\langle \psi, d\psi \rangle, \qquad \mathcal{F} = d\mathcal{A}.$$

Exercise 42 (Berry phase as holonomy)

Prove that $L \to X$ is a complex line bundle. Under the change $\psi \mapsto e^{i\lambda}\psi$, prove that $\mathcal{A} \mapsto \mathcal{A} - d\lambda$ and that $\mathcal{F}$ is unchanged. For a closed curve γ in X, prove that
$$\exp\left(i \int_\gamma \mathcal{A}\right)$$
is the holonomy of the eigenline connection and is independent of the local choice of eigenvector. Interpret it as the geometric phase accumulated during adiabatic transport.

Let P be an invariant symmetric k-linear polynomial on the endomorphism algebra of a fiber of E, where invariant means invariant under simultaneous conjugation. Define [29]

$$P(F_\nabla) = P(F_\nabla, \ldots, F_\nabla) \in \Omega^{2k}(M).$$

Exercise 43 (Chern–Weil theorem)

Prove that
$$dP(F_\nabla) = 0.$$
For two connections ∇^0, ∇^1 on E, prove that

$$[P(F_{\nabla^0})] = [P(F_{\nabla^1})] \in H^{2k}_{\mathrm{dR}}(M).$$

Define the **Chern character form** by

$$\mathrm{ch}(E, \nabla) = \mathrm{tr}\exp\left(\frac{iF_\nabla}{2\pi}\right).$$

When induced connections are understood, suppress them from the notation.

Exercise 44 (Chern character)

Prove that $d\,\mathrm{ch}(E, \nabla) = 0$ and that its de Rham class is independent of ∇. For a second complex vector bundle F with connection, prove that

$$\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F),$$

$$\mathrm{ch}(E \otimes F) = \mathrm{ch}(E) \wedge \mathrm{ch}(F).$$

If M is compact, deduce that the Chern character defines a ring homomorphism

$$\mathrm{ch} : K^0(M) \longrightarrow H^{\mathrm{even}}_{\mathrm{dR}}(M), \qquad H^{\mathrm{even}}_{\mathrm{dR}}(M) = \bigoplus_{j \geq 0} H^{2j}_{\mathrm{dR}}(M).$$

Return to the two-dimensional class A insulator in Exercise 33. For its smooth occupied-state projector P, prove that the Grassmann connection $\nabla = P\,d$ has curvature $F_\nabla = P\,dP \wedge dP$ on E_H^-. Show that the integer classifying the phase is the first Chern number

$$C_1(E_H^-) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \mathrm{tr}\big(P\,dP \wedge dP\big) \in \mathbb{Z},$$

and deduce that a nonzero Chern number obstructs a global smooth frame of occupied Bloch states.

2.6 Chern–Simons theory and TQFT

For two connections on E, set

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \qquad A = \nabla^1 - \nabla^0, \qquad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \mathrm{End}\,E)$, and define

$$\mathrm{CS}_P(\nabla^0, \nabla^1) = k \int_0^1 P(A, F_t, \ldots, F_t)\,dt.$$

This defines $\mathrm{CS}_P(\nabla^0, \nabla^1) \in \Omega^{2k-1}(M)$.

Exercise 45 (Chern–Simons transgression)
Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d\ \mathrm{CS}_P(\nabla^0, \nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M, \mathfrak{g})$, wedge products include matrix multiplication. Define

$$F_A = dA + A \wedge A$$

and

$$\mathrm{CS}(A) = \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

Exercise 46 (Chern–Simons form)
Prove that

$$d\ \mathrm{CS}(A) = \mathrm{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M, define

$$S_{\mathrm{CS}}(A) = \int_M \mathrm{CS}(A).$$

Exercise 47 (Chern–Simons equation)
For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left.\frac{d}{dt}\right|_{t=0} S_{\mathrm{CS}}(A_t) = 2\int_M \mathrm{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \to G$, define

$$A^g = g^{-1}Ag + g^{-1}dg.$$

Assume that G is compact, connected, simple, and simply connected, and normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2}\int_M \mathrm{tr}\big((g^{-1}dg)^3\big) \in \mathbb{Z}.$$

For $k \in \mathbb{R}$, set $S_k = (k/4\pi)S_{\mathrm{CS}}$.

Exercise 48 (Gauge transformation)
Prove that

$$\mathrm{CS}(A^g) - \mathrm{CS}(A) = -d\,\mathrm{tr}(dg\,g^{-1} \wedge A) - \frac{1}{3}\,\mathrm{tr}\big((g^{-1}dg)^3\big).$$

Deduce that $\exp(iS_k)$ *is invariant under all gauge transformations when $k \in \mathbb{Z}$.*

Now restrict to $2+1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2,\mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2,\mathbb{R})), \qquad A^{\pm} = \omega \pm \frac{1}{\ell}e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2,\mathbb{R}) \oplus \mathfrak{sl}(2,\mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \qquad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 49 (Three-dimensional gravity)
Prove that $F_{A^+} = F_{A^-} = 0$ is equivalent to

$$d_\omega e = 0, \qquad F_\omega + \frac{1}{\ell^2}e \wedge e = 0.$$

Assuming that $e : TM \to \mathfrak{sl}(2,\mathbb{R})$ is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.

Define the first-order three-dimensional Einstein–Hilbert functional, up to its conventional overall coefficient, by

$$S_{\mathrm{EH}}(e,\omega) = \int_M \mathrm{tr}\left(e \wedge F_\omega + \frac{1}{3\ell^2}e \wedge e \wedge e\right).$$

Exercise 50 (Gravity as Chern–Simons theory)

Prove that, up to an overall normalization and an exact term,

$$S_{\mathrm{CS}}(A^+) - S_{\mathrm{CS}}(A^-)$$

equals $S_{\mathrm{EH}}(e,\omega)$ *[32, 33].*

This is a local Lie-algebra formulation of gravity in three spacetime dimensions. Changing between Lorentzian and Euclidean signatures changes the relevant real form and reality conditions. It is an analytic continuation, not a statement that four-dimensional gravity is obtained from Chern–Simons theory by Wick rotation.

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\mathrm{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \mathrm{tr}_V(\mathrm{Hol}_\gamma(A)) .$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \ldots, V_r$, set

$$W_L(A) = \prod_{j=1}^{r} W_{V_j}(\gamma_j, A).$$

For a framed link, define the formal Chern–Simons expectation value [30]

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A)\, \mathcal{D}A}{\int e^{iS_k(A)}\, \mathcal{D}A}$$

For normalized $SU(2)$ theory at integer level $k \geq 1$, set

$$q = \exp\left(\frac{2\pi i}{k+2}\right).$$

Exercise 51 (Wilson loops and the Jones polynomial)

Prove that the expectation value depends on the chosen framing. Let L_+*,* L_-*, and* L_0 *differ only by a positive crossing, a negative crossing, and the oriented smoothing. Assume that, after the standard framing normalization, the Wilson-loop expectation values satisfy*

$$q^{-1}\langle W_{L_+}\rangle_k - q\langle W_{L_-}\rangle_k = \left(q^{1/2} - q^{-1/2}\right)\langle W_{L_0}\rangle_k$$

and $\langle W_{\bigcirc}\rangle_k = 1$*. Prove that these relations determine the Jones polynomial of an oriented link.*

Let $\mathrm{Bord}_3^{\mathrm{or}}$ be the symmetric monoidal category whose objects are closed oriented surfaces, whose morphisms are oriented three-dimensional bordisms modulo diffeomorphism, and whose tensor product is disjoint union. A **three-dimensional oriented TQFT** is a symmetric monoidal functor

$$Z : \mathrm{Bord}_3^{\mathrm{or}} \longrightarrow \mathrm{Vect}_{\mathbb{C}} .$$

Exercise 52 (Reshetikhin–Turaev TQFT)

Let $\mathcal{C}$ be a modular braided fusion category. Prove that the Reshetikhin–Turaev construction assigns finite-dimensional state spaces to closed oriented surfaces and compatible linear maps to bordisms, defining a three-dimensional oriented TQFT [31]. Prove that labeled Wilson lines induce the braid-group actions determined by the F- and R-symbols of $\mathcal{C}$. For the $SU(2)_k$ category, relate the resulting link invariant to the Chern–Simons Wilson-loop skein relation above.

3 Analysis

3.1 Norms, operator exponentials, and linear evolution

A **norm** on a complex vector space H is a function $\|\cdot\| : H \to [0, \infty)$ that is definite, homogeneous, and satisfies the triangle inequality. A normed space is a **Banach space** if every Cauchy sequence converges. A **Hilbert space** is a complex vector space with a Hermitian inner product $\langle \cdot, \cdot \rangle$ whose norm $\|x\| = \sqrt{\langle x, x \rangle}$ is complete. A linear map $A : H \to K$ is **bounded** if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty.$$

Write $\mathcal{B}(H, K)$ for the normed space of bounded operators and $\mathcal{B}(H) = \mathcal{B}(H, H)$.

For $A \in \mathcal{B}(H)$, define the operator exponential by the formal series

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

whenever the series converges in operator norm.

Exercise 53 (Cauchy–Schwarz inequality)

Prove that every inner-product space satisfies

$$|\langle x, y \rangle| \le \|x\| \|y\|.$$

Deduce the triangle inequality for the induced norm. Prove that $\mathcal{B}(H, K)$ is complete when K is complete and that the exponential series converges in $\mathcal{B}(H)$ for every bounded A.

Exercise 54 (Operator exponentials and linear evolution)

Prove that e^{tA} is differentiable in operator norm and that

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

Deduce that the unique solution of

$$\dot{x}(t) = Ax(t), \qquad x(0) = x_0,$$

is $x(t) = e^{tA}x_0$. In finite dimensions use the Jordan decomposition

$$A = S + N, \qquad [S, N] = 0,$$

with S semisimple and N nilpotent, to prove

$$e^{tA} = e^{tS}e^{tN}.$$

Show that on the generalized λ-eigenspace every component of a solution is a linear combination of terms $t^j e^{\lambda t}$.

3.2 Hilbert-space operator theory

An **adjoint** of $A \in \mathcal{B}(H, K)$ is an operator $A^* \in \mathcal{B}(K, H)$ satisfying

$$\langle Ax, y \rangle = \langle x, A^*y \rangle.$$

An operator $A \in \mathcal{B}(H)$ is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle x, Ax \rangle \geq 0$ for every x, and **unitary** if $A^*A = AA^* = 1$.

Exercise 55 (Adjoints of bounded operators)
Prove that every bounded operator between Hilbert spaces has a unique adjoint and establish the identities

$$(AB)^* = B^*A^*, \qquad \|A^*A\| = \|A\|^2.$$

3.3 Compact operators and the Fredholm alternative

A bounded operator $K : H \to L$ between Hilbert spaces is **compact** if the image of the closed unit ball has compact closure. Equivalently, for every bounded sequence (x_n) in H, the sequence (Kx_n) has a convergent subsequence. Write $\mathcal{K}(H, L)$ for the compact operators and $\mathcal{K}(H) = \mathcal{K}(H, H)$.

Exercise 56 (Compact operators)
Prove the sequential characterization of compactness above. Prove that every finite-rank operator is compact, that $\mathcal{K}(H, L)$ is closed in the operator norm, and that

$$A \in \mathcal{B}(L, M), \quad K \in \mathcal{K}(H, L), \quad B \in \mathcal{B}(N, H) \quad \Longrightarrow \quad AKB \in \mathcal{K}(N, M).$$

Deduce that $\mathcal{K}(H)$ is a closed two-sided ideal in $\mathcal{B}(H)$. Prove that K^* is compact whenever K is compact and that, on a Hilbert space, every compact operator is an operator-norm limit of finite-rank operators.

A bounded operator $T : H \to L$ is **Fredholm** if $\ker T$ and $L/\operatorname{im} T$ are finite dimensional and $\operatorname{im} T$ is closed. Its **Fredholm index** is

$$\operatorname{ind}(T) = \dim \ker T - \dim(L/\operatorname{im} T).$$

Exercise 57 (Fredholm alternative)

Let $K \in \mathcal{K}(H)$ and $\lambda \in \mathbb{C} \setminus \{0\}$. Prove that $\lambda 1 - K$ has closed range and that [11]

$$\dim \ker(\lambda 1 - K) = \dim \ker(\overline{\lambda} 1 - K^*) < \infty.$$

Using

$$\operatorname{im}(\lambda 1 - K)^{\perp} = \ker(\overline{\lambda} 1 - K^*),$$

prove that the following are equivalent:

- $\lambda 1 - K$ *is injective.*
- $\lambda 1 - K$ *is surjective.*
- $\lambda 1 - K$ *is invertible.*

Prove more generally that $(\lambda 1 - K)x = y$ is solvable if and only if y is orthogonal to $\ker(\overline{\lambda} 1 - K^*)$. Deduce that $\lambda 1 - K$ is Fredholm of index zero. Finally, prove that every nonzero point of the spectrum of a compact operator is an eigenvalue of finite multiplicity and that zero is the only possible accumulation point of the spectrum.

3.4 Quantum states and uncertainty

A finite-dimensional **quantum state** is a positive operator ρ with $\operatorname{tr}(\rho) = 1$. A unit vector ψ defines the pure state $\rho_\psi = |\psi\rangle\langle\psi|$. For a self-adjoint observable A, define

$$\langle A \rangle_\rho = \operatorname{tr}(\rho A), \qquad \operatorname{Var}_\rho(A) = \operatorname{tr}\big(\rho(A - \langle A \rangle_\rho 1)^2\big).$$

For $\rho = \rho_\psi$, use the subscripts ψ instead of ρ.

Exercise 58 (Robertson uncertainty inequality)

For self-adjoint $A, B \in \mathcal{B}(H)$ and a unit vector ψ, prove [23]

$$\operatorname{Var}_\psi(A) \operatorname{Var}_\psi(B) \geq \frac{1}{4} \left| \langle \psi, [A, B]\psi \rangle \right|^2, \qquad [A, B] = AB - BA.$$

Determine the equality condition.

For a self-adjoint H, write $U_t = e^{-itH}$.

Exercise 59 (Conservation from commutators)

Let A, H be self-adjoint. Prove that U_t is unitary and that [17]

$$\frac{d}{dt} \langle U_t \psi, A U_t \psi \rangle = -i \langle U_t \psi, [H, A] U_t \psi \rangle.$$

Deduce that $[H, A] = 0$ if and only if the expectation of A is constant under this evolution for every initial vector.

Exercise 60 (No-cloning theorem)

Let $b \in H$ be a unit vector and suppose that a unitary $C : H \otimes H \to H \otimes H$ satisfies $C(v \otimes b) = v \otimes v$ and $C(w \otimes b) = w \otimes w$ for unit vectors v, w. Prove that [22]

$$\langle v, w \rangle = \langle v, w \rangle^2.$$

Deduce that no unitary map clones every unit vector.

3.5 Schmidt decomposition and entanglement

For finite-dimensional Hilbert spaces H_A, H_B and $\psi \in H_A \otimes H_B$, define

$$T_\psi : H_A^* \to H_B, \qquad T_\psi(\phi) = (\phi \otimes 1)(\psi).$$

A **Schmidt decomposition** of ψ is an expression

$$\psi = \sum_{i=1}^{r} s_i a_i \otimes b_i.$$

where (a_i) and (b_i) are orthonormal families and $s_i > 0$; r is the **Schmidt rank**. For a unit vector ψ, define the reduced density operators by

$$\rho_A = \mathrm{Tr}_{H_B}(|\psi\rangle\langle\psi|), \qquad \rho_B = \mathrm{Tr}_{H_A}(|\psi\rangle\langle\psi|).$$

Exercise 61 (Schmidt decomposition and entanglement)

Use the singular-value decomposition of T_ψ to prove that every ψ has a Schmidt decomposition. Prove that the nonzero eigenvalues of ρ_A and ρ_B are the numbers s_i^2, with the same multiplicities. Prove that ψ is separable, meaning $\psi = a \otimes b$, exactly when its Schmidt rank is one. Deduce that the entanglement entropy is [17]

$$S(\psi) = -\operatorname{Tr}(\rho_A \log \rho_A) = -\sum_i s_i^2 \log s_i^2,$$

with the convention $0 \log 0 = 0$.

3.6 CHSH, Gisin, and Tsirelson

Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$. Their **CHSH operator** is

$$\mathcal{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Exercise 62 (Gisin's theorem for two qubits)

Prove that every classical local-hidden-variable assignment satisfies $|\langle \mathcal{C} \rangle| \leq 2$. For the two-qubit state in Schmidt form

$$\psi_\theta = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle, \qquad 0 \leq \theta \leq \frac{\pi}{4},$$

optimize over projective binary observables and prove that the maximal CHSH value is [18]

$$2\sqrt{1+\sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates a CHSH inequality. For $\theta = \pi/4$, recover the value $2\sqrt{2}$.

A **bipartite binary quantum correlation matrix** has entries

$$c_{ij} = \langle \psi, (A_i \otimes B_j)\psi \rangle,$$

where ψ is a unit vector. The operators A_i and B_j are self-adjoint unitaries on finite-dimensional Hilbert spaces.

For a real matrix $M = (M_{ij})$, define

$$\beta_{\mathrm{C}}(M) = \max_{\varepsilon_i, \delta_j \in \{-1,1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_{\mathrm{Q}}(M) = \sup_{c} \left| \sum_{i,j} M_{ij} c_{ij} \right|,$$

where the supremum is over bipartite binary quantum correlation matrices.

Exercise 63 (Tsirelson's representation theorem and bound)

Prove that a finite real matrix $c = (c_{ij})$ is a bipartite binary quantum correlation matrix if and only if there are unit vectors x_i, y_j in a real Hilbert space such that [19]

$$c_{ij} = \langle x_i, y_j \rangle.$$

Deduce that

$$\beta_{\mathrm{Q}}(M) = \sup_{\|x_i\| = \|y_j\| = 1} \left| \sum_{i,j} M_{ij} \langle x_i, y_j \rangle \right|.$$

For the CHSH operator, prove the Tsirelson bound $\|\mathcal{C}\| \leq 2\sqrt{2}$ and exhibit observables and a state attaining equality.

3.7 Grothendieck inequality

The **real Grothendieck constant** $K_G^{\mathbb{R}}$ is the least constant for which

$$\beta_{\mathrm{Q}}(M) \leq K_G^{\mathbb{R}} \beta_{\mathrm{C}}(M)$$

holds for every real matrix M.

Exercise 64 (Grothendieck's inequality)
Prove that $K_G^{\mathbb{R}} < \infty$ [20, 21]. For

$$M_{\mathrm{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

compute $\beta_{\mathrm{C}}(M_{\mathrm{CHSH}})$ directly. Choose four unit vectors in $\mathbb{R}^2$ that attain

$$\beta_{\mathrm{Q}}(M_{\mathrm{CHSH}}) = 2\sqrt{2}$$

and verify the ratio $\sqrt{2}$. Deduce that $\sqrt{2} \leq K_G^{\mathbb{R}}$ and interpret the inequality as a dimension-independent bound on quantum advantage for bipartite binary correlation Bell inequalities.

3.8 Von Neumann algebras and subfactors

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$-subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$ [12]. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \to T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta \rangle \longrightarrow \langle T\xi, \eta \rangle$$

for every $\xi, \eta \in H$.

Exercise 65 (Bicommutant theorem)
Let $A \subseteq \mathcal{B}(H)$ be a unital $*$-subalgebra. Prove that

$$A'' = \overline{A}^{\mathrm{SOT}} = \overline{A}^{\mathrm{WOT}}.$$

The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$. A positive functional or positive map is **normal** if it preserves suprema of bounded increasing nets of positive elements, and a positive functional ψ is **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type** II_1 **factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \longrightarrow \mathbb{C}, \qquad \tau(ab) = \tau(ba), \qquad \tau(1) = 1.$$

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \qquad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 66 (Projection dimension)

Let M be a type II_1 factor with normalized trace τ. Prove that

$$p \sim q \quad \Longrightarrow \quad \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \quad \Longleftrightarrow \quad \tau(p) \le \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. For a linear map $\Phi : M \to N$, define its matrix amplification by $\Phi_n([x_{ij}]) = [\Phi(x_{ij})]$; the map is **completely positive** if every Φ_n is positive. Let $N \subseteq M$ be finite von Neumann algebras with a common normalized trace τ. A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \to N$ such that

$$\tau \circ E_N = \tau, \qquad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^* x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 67 (Conditional expectation and Jones projection)

Prove that there is a unique trace-preserving conditional expectation $E_N : M \to N$. Prove that E_N extends to the orthogonal projection

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \qquad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N. For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N \big(pL^2(N, \tau)^k\big) = \sum_{i=1}^{k} \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M, \tau)$ is isomorphic as a right N-module to such a module; then

$$[M : N] = \dim_N L^2(M, \tau).$$

Exercise 68 (Jones basic construction)
Let $N \subseteq M$ be an inclusion of type II_1 *factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [13]*

$$\tau_1(xe_N y) = \frac{1}{[M:N]}\tau(xy)$$

for $x, y \in M$.

The **Jones tower** is the sequence obtained by iterating the basic construction:

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots .$$

Let e_i be the Jones projection used to construct M_i from M_{i-1}, and set

$$\delta = \sqrt{[M:N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \qquad U_iU_j = U_jU_i \quad (|i-j| \geq 2), \qquad U_iU_{i\pm1}U_i = U_i.$$

Exercise 69 (Temperley–Lieb relations)
Prove that the Jones projections satisfy [16, 14]

$$e_i^2 = e_i = e_i^*, \qquad e_ie_j = e_je_i \quad (|i-j| \geq 2),$$
$$e_ie_{i\pm1}e_i = \delta^{-2}e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define

$$\sigma_i = q\,1 - U_i.$$

These operators reconnect the Temperley–Lieb construction with the braid groups introduced in Algebra.

Exercise 70 (Braid representation)
Prove that σ_i is invertible with inverse $q^{-1}1 - U_i$, and prove that

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},$$
$$\sigma_i\sigma_j = \sigma_j\sigma_i \qquad (|i-j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [15].

A **Markov trace** on the Temperley–Lieb tower is a compatible family of positive normalized traces satisfying

$$\mathrm{tr}(xU_i) = \delta^{-1}\,\mathrm{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators. Define

$$P_0(\delta) = 1, \qquad P_1(\delta) = \delta, \qquad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Exercise 71 (Markov trace and discrete index values)

Prove that the normalized traces on the Jones tower restrict to a Markov trace. Prove that, when $\sin\theta \neq 0$*,*

$$P_n(2\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$*, then*

$$\delta = 2\cos\left(\frac{\pi}{m}\right)$$

for some integer $m \geq 3$*.*

Exercise 72 (Jones index theorem)

Let $N \subseteq M$ *be an inclusion of type* II_1 *factors of finite Jones index. Prove that [14, 13]*

$$[M : N] \in \left\{4\cos^2\left(\frac{\pi}{n}\right) : n \geq 3\right\} \cup [4, \infty).$$

For

$$[M : N] = 4\cos^2\left(\frac{\pi}{k+2}\right), \qquad \delta = \sqrt{[M : N]},$$

prove that the Jones projections determine a representation of the Temperley–Lieb algebra $\mathrm{TL}_n(\delta)$*, and hence a braid-group representation. Show that the Markov trace of a braid closure satisfies the Jones skein relation. Relate the parameter*

$$\delta = 2\cos\left(\frac{\pi}{k+2}\right)$$

to the $SU(2)_k$ *fusion and braiding data from Algebra and to the Wilson-loop invariant of Chern–Simons theory from Geometry.*

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